On my honor I have never given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

Yunus Emre Ertunç　　19.04.2021

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Q1) $t \in (0,13)$　　$f(t) = \sum_{k=-\infty}^{\infty} (t-2k+3)\, \chi_{(0,1)}(t-2k)$

a)

$\vdots$

$k=-2$　$(t+7)\, \chi_{(0,1)}(t+4)$

$k=-1$　$(t+5)\, \chi_{(0,1)}(t+2)$

$k=0$　$(t+3)\, \chi_{(0,1)}(t)$　$\propto$.

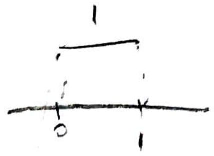
$k=1$　$(t+1)\, \chi_{(0,1)}(t-2)$

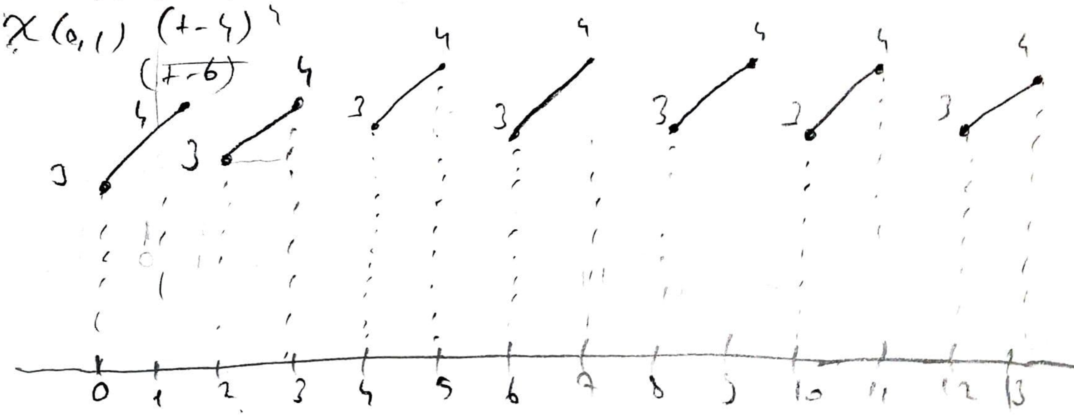
$k=2$　$(t-1)\, \chi_{(0,1)}(t-4)$

$t-3$　　$(t-6)$

$\vdots$

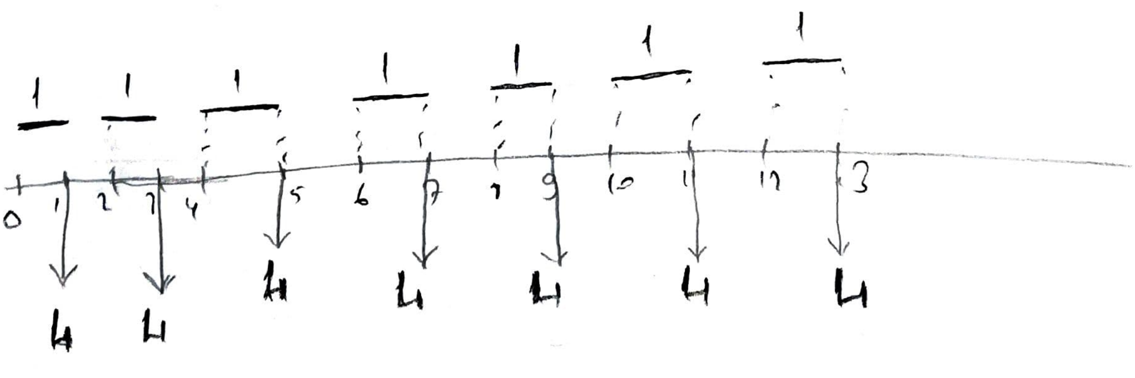
$\chi_{(0,1)}(t) =$





b)

$\frac{d\,f(t)}{dt}$



Yunus Emre
Ertunç

2) $$\lim_{a \to \infty} \int_{-5}^{10} \underbrace{\frac{t^2+20}{e^t(t-3)}}_{y(t)} \sin(a(t-3))\,dt$$

$$\lim_{\omega \to \infty} f(t) \sin(\omega t) = 0$$ because of R-L

$$\lim_{a \to \infty} \int_{-5}^{10} y(t) \sin(a(t-3))\,dt$$

Answer is 0

R-L